

Houru Jiang

Email: tifeng236@gmail.com

juliourias@stu2023.jnu.edu.cn

Phone: +86 18938291235

Guangzhou, China

Website: simontifeng.github.io

Research Profile

I work across **recommendation, graph learning, large language models, time series, and interpretable AI**. My research uses mathematical and model-agnostic methods – including compression, entropy, causal intervention, and uncertainty estimation – to improve learning efficiency and diagnose model behaviour. I am particularly interested in efficient LLM training and architectures, including principled pre-training and post-training methods and alternatives such as linear attention.

Core Interests:

Recommendation and graph learning; large language models and efficient training; time-series learning; interpretable and trustworthy AI; mathematical methods for data selection, pre-training, and post-training; linear attention and other efficient LLM architectures.

Selected Manuscripts and Submissions

• An Information-Theoretic Evaluation Framework for Benchmark and Model Diagnosis in Knowledge Tracing

Houru Jiang*, Zixi Wang*, Tengteng Cheng, Xueyi Li, Mingliang Hou, Jiaqi Zheng, Renqiang Luo, Teng Guo, Zitao Liu
Accepted as a poster at NeurIPS 2026 Main Track. Co-First Author.

• Residualizing DKT with Symbolic Sequence Priors for Knowledge Tracing

Zixi Wang*, Houru Jiang*, Tengteng Cheng, Xueyi Li, Mingliang Hou, Zitao Liu
Submitted to AISTATS 2027. Co-First Author.

• When Target Accuracy Hides Semantic Interface Drift in Vision–Language Adaptation

Houru Jiang, Zixi Wang, Mingliang Hou, Xueyi Li, Teng Guo, Qianru Liang, Zitao Liu
Under review at ICLR 2027. First Author.

• Factorized Initialization for Few-Step Adaptation in Time-Series Forecasting

Houru Jiang, Zixi Wang, Mingliang Hou, Dan Lin, Qianru Liang
Under review at ICLR 2027. First Author.

• Predicted Links as Messages: Separating Formation from Propagation in Temporal Graphs

Houru Jiang, Zixi Wang, Mingliang Hou, Tengteng Cheng, Teng Guo, Qianru Liang, Zitao Liu
Under review at ICLR 2027. First Author.

• HB++: Topology-Conditioned Propagation Bases for Shallow Graph Learning

Houru Jiang, Zixi Wang, Mingliang Hou, Tengteng Cheng, Zitao Liu, Teng Guo, Jiaqi Zheng
Under review at WSDM 2027. First Author.

* Equal contribution / co-first authors.

Research Experience

Research Assistant

Nov 2024 – Sep 2025

Professor Hao Chen, Macau University of Science and Technology (Remote)

- Conducted research on knowledge graphs, graph representation learning, recommendation systems, and knowledge-enhanced retrieval.
- Analyzed how structured knowledge and retrieval signals can support recommendation and information-access systems, contributing to data organization, method implementation, and experimental validation.

Research Assistant

May 2025 – Present

Associate Professor Mingliang Hou
Jinan University

- Designed CDIS, a CPU-only instruction-data selector that couples compression ratio and token entropy for low-cost, interpretable LLM data selection.
- Developed an information-theoretic framework for localized benchmark and model diagnosis in knowledge tracing.
- Built causal and probabilistic auditing methods to test whether graph-based models genuinely utilize their input graph structures.
- Contributed to a robotic-manipulation project through system evaluation, experimental analysis, and reproducible comparison studies.

Education

Jinan University

Guangzhou, China

Bachelor of Information Management and Information Systems, Department of Mathematics

Expected Graduation: Jun 2027

GPA: 3.34/5.00; Average: 83.4/100

Relevant Coursework

Mathematics: Mathematical Analysis I–III, Advanced Algebra I–II, Probability and Statistics, Discrete Mathematics, Operations Research

Computer Science: Data Structures, Object-Oriented Programming, Database Systems, Computer Networks

Artificial Intelligence: Computational Intelligence, Advanced Artificial Intelligence

Technical Skills

Programming and Scientific Computing: Python, MATLAB, SQL, Java, C, Pandas, NumPy, Matplotlib, Seaborn

Machine Learning and LLM Systems: PyTorch, Scikit-Learn, Transformers, Hugging Face, Qwen, DeepSeek, OpenAI API, Ollama, Stable-Baselines3

Tools: Linux, Git, Jupyter, LaTeX

Honors and Awards

- Jinan University Third-Class Outstanding Student Scholarship
- Mathematical Contest in Modeling (MCM/ICM), Successful Participant (2024)
- Mathematical Contest in Modeling (MCM/ICM), Honorable Mention (2025)

Language Proficiency

IELTS: Overall 7.0

Listening 7.5 | Reading 7.5 | Speaking 6.0 | Writing 6.0